

Slutsky Decomposition of IE and SE, Slutsky Demand, Index Numbers

JEB104 Microeconomics I
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Prague, IES FSV UK, April 6, 2020

Slutsky Decomposition of IE and SE

Change in price → change in relative prices, change in real income

- ➊ Income effect: effect of change in real income
- ➋ Substitution effect: effect of change in relative prices
 - Hicksian decomposition: constant utility
 - Slutsky decomposition: constant purchasing power

Previously – Hicksian decomposition of total price effect of a price change into income and substitution effects

- We know map of indifference curves
- Substitution effect: change in demand that is caused by a pure change in relative prices, while keeping the original level of utility (movement along the original indifference curve to a point where $|MRS_{21}| = \text{new price ratio}$)
- based on keeping **constant utility**

Today

- Problem with Hicksian decomposition is that preferences are unobservable
- Can we study price changes in a greater detail when only CHOICES are observable?
- Decomposition into SE and IE can be based on constant purchasing power (rather than on constant utility)
 - Builds on revealed preferences covered last time

This slide just reminds you that:

- If price changes, then this change leads to a change in relative prices (ratio in which consumer could substitute one good for another in the market) and also to a change in real income (how many units of goods consumer could buy – not only of a good whose price change)
- There are two different approaches how to analyze a change in demand due to a price change in a greater detail
- We covered Hicksian decomposition several weeks ago, we will discuss Slutsky decomposition now.

Constant purchasing power – what does it mean to keep purchasing power constant?

- After the price change (i.e. under new relative prices), consumer can just afford the bundle which he was originally purchasing.

Slutsky Income Compensation

Constant purchasing power

- after the price change, consumer may buy the same bundle as before the price change, given that his entire income is spent.

Before price change: income m^0 , prices $\mathbf{p}^0 \Rightarrow$ optimum $\mathbf{x}^{0*}$, $\sum_{j=1}^n p_j^0 x_j^0 = m^0$

Suppose that price of good 1 increases: $p_1^0 < p_1^1$

After price change: income m^0 , prices $\mathbf{p}^1 \Rightarrow$ optimum $\mathbf{x}^{1*}$, $\sum_{j=1}^n p_j^1 x_j^1 = m^0$

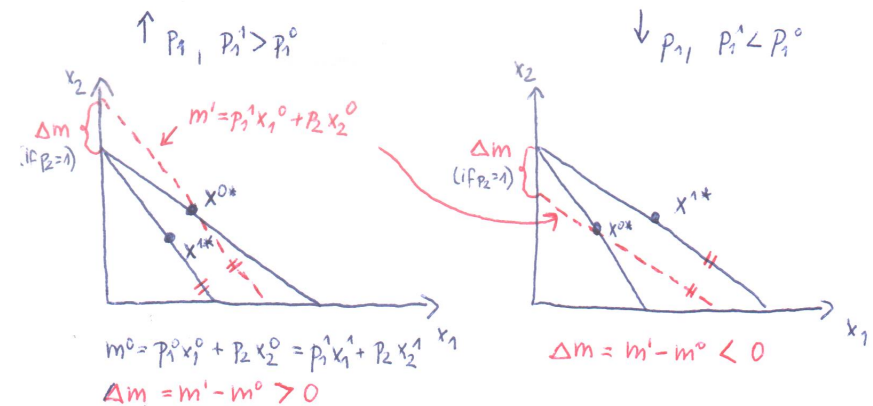
$\mathbf{x}^0$ is unaffordable due to $\uparrow p$ at the initial income m^0 .

To keep the same purchasing power after the price change, the consumer has to have a compensated income with which he could afford the original bundle $\mathbf{x}^{0*}$ under new prices $\mathbf{p}^1$, so $m' = \sum_j p_j^1 x_j^0$.

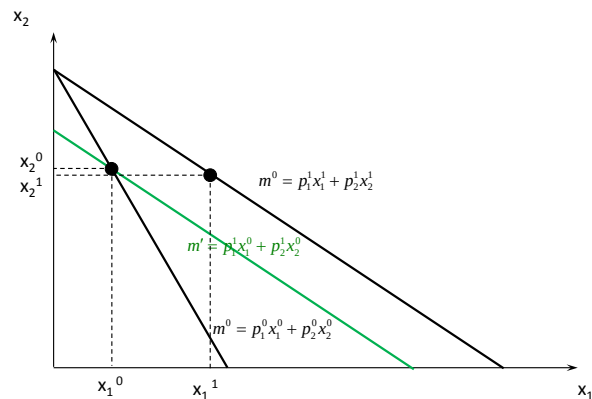
Slutsky income compensation $\Delta m = m' - m^0 = \sum_j (p_j^1 - p_j^0) x_j^0$

- the amount of money given ($\uparrow p$) or taken away ($\downarrow p$) that makes the original bundle consumed just affordable under the new prices.
- the change in income after the price change that keeps consumer's purchasing power constant.

Pictures below show both cases: $\uparrow p$ and $\downarrow p$



Slutsky Income Compensation



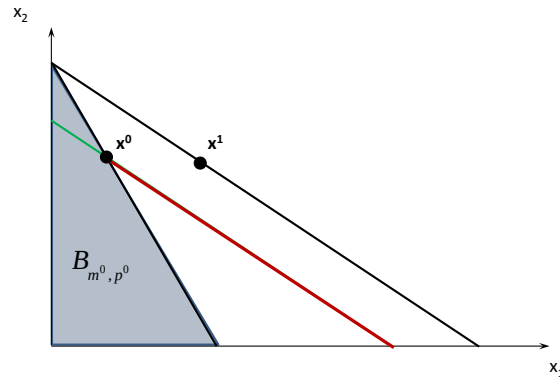
Which bundle $\mathbf{x}'$ will be chosen if the consumer faces the green budget constraint?

The picture on the slide is equivalent to the picture on the right panel above.

To decompose change in demand due to change in price, hence movement from $\mathbf{x}^{0*}$ to $\mathbf{x}^{1*}$, to substitution and income effect, we need to see what the consumer would choose when facing the compensated budget, i.e. having income m' and facing new prices $\mathbf{p}^1$ (green budget line)

So in contrast to Hicksian decomposition, we do not need to know the indifference curves and their shape. We need to know what bundle is chosen under m' and $\mathbf{p}^1$.

Slutsky Decomposition of IE and SE



x^0 directly revealed preferred to all bundles in B_{p^0, m^0}

x' located on the **red part** of the budget line.

- the original bundle x^0 cannot be chosen due to strict convexity of preferences
- the bundles in B_{m^0, p^0} cannot be chosen (WARP)

Could we eliminate any points on the budget line which would not be chosen by a rational consumer when he has m' and faces p^1 ?

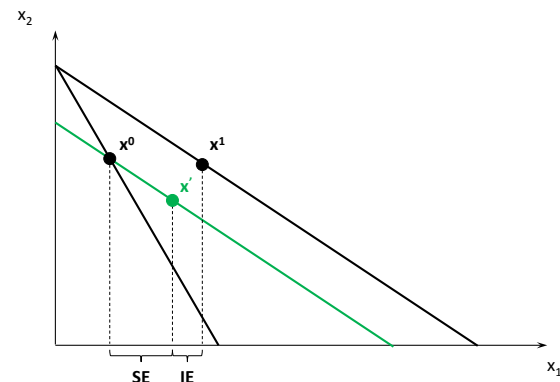
Yes, let's apply the concept of revealed preferences which we learned last time.

We assume that underlying preferences are strictly convex (the assumption was also imposed when we discussed revealed preferences), because then optimal bundle is unique for any specific price and income combination.

- Optimal bundle must lie on the green budget line (all income will be spent, this is trivial).
- Optimal bundle cannot be x^0 , because x^0 is optimal for a different price-income combination.
- So our optimal bundle x' will be preferred to x^0 .
- As x^0 is directly revealed preferred to all bundles in the B_{m^0, p^0} , x^0 is affordable, no bundles in B_{m^0, p^0} can be optimal. This follows from the WARP: if x is directly revealed to y , then y can never be directly revealed to x (x is our x^0 and y is a bundle on the green part of the budget line)
- Hence, bundles on the green part of the budget line cannot be chosen.

Only bundles on red part of the budget line can be chosen.

Slutsky Decomposition of IE and SE



Substitution effect: $x^0 \rightarrow x'$, Income effect: $x' \rightarrow x^1$

Substitution effect: opposite sign to the price change

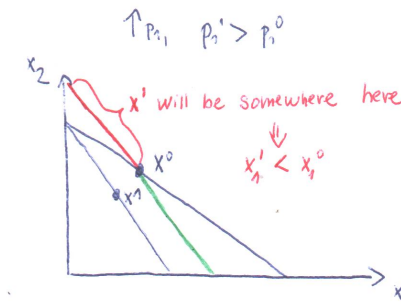
$$\left. \begin{array}{l} \Delta p_i < 0 : x'_i - x_i^0 = \Delta x_i \left| \sum_{j=1}^n p_j x_j^0 = m \right. > 0 \\ \Delta p_i > 0 : x'_i - x_i^0 = \Delta x_i \left| \sum_{j=1}^n p_j x_j^0 = m \right. < 0 \end{array} \right\} \Rightarrow \frac{\Delta x_i \left| \sum_{j=1}^n p_j x_j^0 = m \right.}{\Delta p_i} < 0$$

This picture shows where x' may lie. In this case, $x_1^0 < x_1' < x_1^1$ (x_1 is normal good), we can have also a situation that $x_1^0 < x_1' < x_1^1$ (x_1 is an inferior good). Nevertheless, we will always have:

$$x_1^0 < x_1'$$

So if price of good i decreases ($\Delta p_i = p_i^1 - p_i^0 < 0$), the change in demand due to Slutsky substitution effect will be positive, i.e. $x_1' - x_1^0 > 0$.

If price of good i increases, the change in demand due to Slutsky substitution effect will be negative, i.e. $x_1' - x_1^0 < 0$, as shown on the picture below.



So as in Hicksian decomposition, own substitution effect has always an opposite sign to own price change.

Slutsky Decomposition of IE and SE

Income effect

Change in real income:

$$\Delta m_{real} = m^0 - m' = \sum_{j=1}^n p_j^1 x_j^1 - \sum_{j=1}^n p_j^1 x_j^0 = \sum_{j=1}^n p_j^0 x_j^0 - \sum_{j=1}^n p_j^1 x_j^0 = \sum_{j=1}^n (p_j^0 - p_j^1) x_j^0 = - \sum_{j=1}^n \Delta p_j x_j^0$$

$$\Delta p_i > 0 \Rightarrow \Delta m_{real} < 0, \Delta p_i < 0 \Rightarrow \Delta m_{real} > 0$$

$$IE = \Delta m_{real} \frac{\Delta x_i(p, m)}{\Delta m}$$

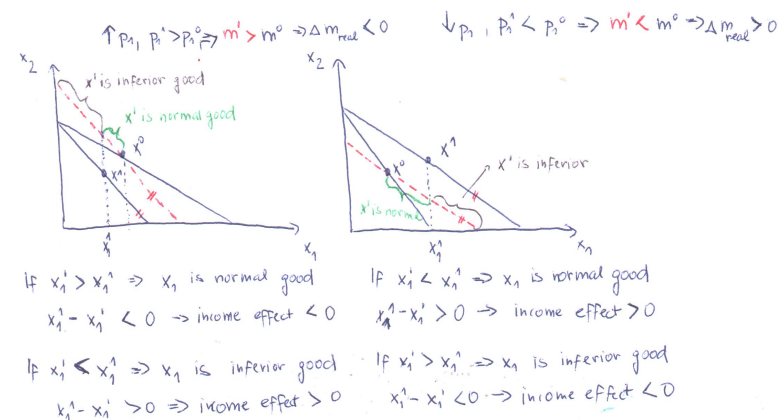
Normal good: IE has the opposite sign to the price change

Inferior good: IE has the same sign as the price change

Remember that the income effect is a change in demand due to a change in real income

We can do the second step because $\sum_{j=1}^n p_j^1 x_j^1 = m^0 = \sum_{j=1}^n p_j^0 x_j^0$ Δm_{real} is opposite to price change = negative of Slutsky income compensation

The pictures below show 4 cases which may arise (increase of price on the left panel, decrease of price on the right panel), while each case shows where x' can be located given the good is normal or inferior. Remember that from previous slide we know that $x_1' < x_1^0$ in the case of price increase and $x_1' > x_1^0$ in the case of price decrease.



Slutsky Equation

$$(x_j^1 - x_j^0) = (x_j' - x_j^0) + (x_j^1 - x_j') \cdot \frac{1}{\Delta p_i}$$

$$\frac{(x_j^1 - x_j^0)}{\Delta p_i} = \frac{(x_j' - x_j^0)}{\Delta p_i} + \frac{(x_j^1 - x_j')}{\Delta p_i}$$

Because $\Delta m_{real} = -\Delta p_i x_i^0 \Rightarrow \Delta p_i = -\frac{\Delta m_{real}}{x_i^0}$, we have:

$$\frac{(x_j^1 - x_j^0)}{\Delta p_i} = \frac{(x_j' - x_j^0)}{\Delta p_i} + (-x_i^0) \frac{(x_j^1 - x_j')}{\Delta m}$$

Rewriting with Δx_j for change in x_j ,

$$\frac{\Delta x_j}{\Delta p_i} \Big|_{\Delta m=0} = \frac{\Delta x_j}{\Delta p_i} \Big|_{m=px^0} - x_i^0 \frac{\Delta x_j}{\Delta m} \Big|_{p=p^1}$$

Now we can derive modified Slutsky equation for Slutsky decomposition of SE and IE.

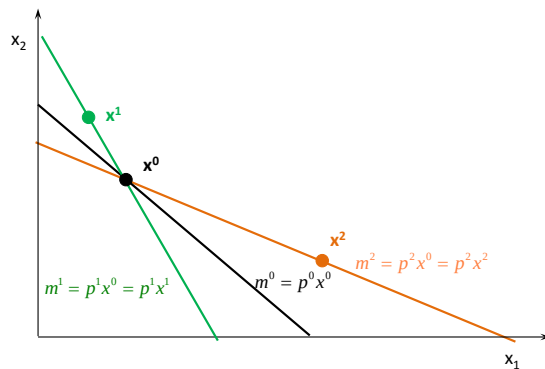
- Left-hand side – total effect – change in Walrasian demand due to price change
- Right-hand side – substitution effect + income effect
- Divided by change in price – rate of change
- Instead of Δm_{real} we can write Δm , because demand changes by the same rate with changes in m and m_{real}

The equation at the bottom:

- $\Delta m = 0$ means we do not compensate the consumer, it is just the Walrasian demand, which shows change in demand with change of price while consumer has a certain level of income.
- $m = px^0$ means constant purchasing power – income is adjusted in such a way that original bundle x^0 is just affordable.
- $p = p^1$ – income effect under new-prices ratio

For Hicksian decomposition, the first term on the right-hand side represented the slope of Hicksian demand. This cannot be the case for this Slutsky equation. In this case, this term represents the slope of **Slutsky demand**.

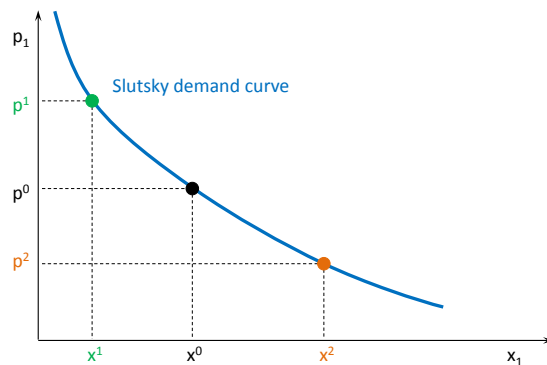
Slutsky Demand Curve



- 1 Before Δp . Optimal x^0 ($m^0 = p_1^0 x_1^0 + p_2^0 x_2^0$)
- 2 $p_1^0 \nearrow p_1^1$, but income adjusted: $m' = p_1^1 x_1^0 + p_2^0 x_2^0 > m^0$. New optimum x^1
- 3 $p_1^0 \searrow p_1^2$, but income adjusted: $m' = p_1^2 x_1^0 + p_2^0 x_2^0 < m^0$. New optimum x^2

Slutsky Demand Curve

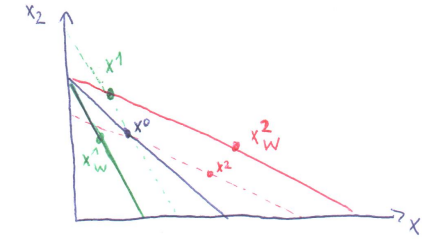
Slutsky demand $S(p_1, \bar{p}_2, m = px^0) = S(p, m = px^0)$



Always downward-sloping (substitution effect has opposite sign to Δp).

This slide shows how you can derive Slutsky demand. There is one Slutsky demand (for a given good) for each initial optimal bundle x^0 . Slutsky demand shows you what other bundles are demanded with change of one price given that the initial bundle is always affordable (so your income is always adjusted for each price change such that the initial bundle is just affordable after the price change).

The picture shows you also how Walrasian demand changes with these price changes.

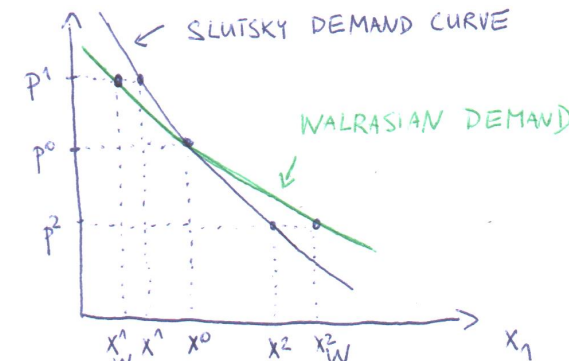


From the figure above, we can derive the Slutsky demand – corresponding prices are shown at vertical axis.

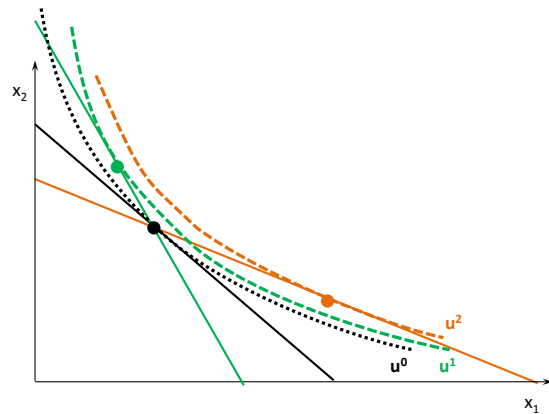
Slutsky demand is always downward-sloping (follows from WARP)

- This is the same as with Hicksian demand

The picture below shows both, Slutsky demand and Walrasian demand. In this case, Walrasian demand is flatter. It is due to the fact, that the good x_1 is normal – this can be seen from the picture above and comparing x_{1w}^1 and x_1^1 ($x_{1w}^1 < x_1^1$) and x_{1w}^2 and x_1^2 ($x_{1w}^2 > x_1^2$).



Slutsky Demand Curve



After the price change and Slutsky income compensation:

- always at least as high utility as before the price change (does not matter whether price increases or decreases)

Notice that points on Slutsky demand will always be on a higher indifference curve than the original indifference curve (i.e. curve passing through the point x^0) regardless of price change

- Graphically - the compensated budget constraint will pass through the current indifference curve
- Therefore, new chosen bundle has to lie on a higher IC

Comparison with the other two demand curves:

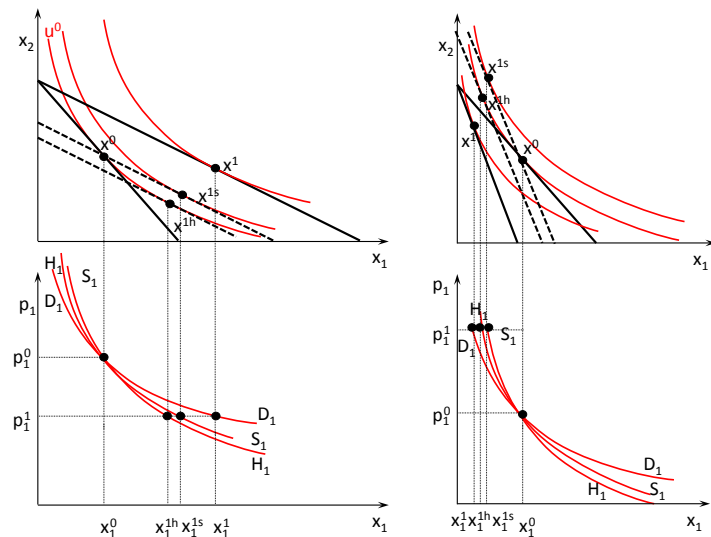
- Hicksian demand: the same utility
- Walrasian demand: higher utility when price decreases, lower utility when price increases

What does this mean for the three demand curves and their relative positions (flatter/steeper)?

- For normal good (left to right / low to high): $p \uparrow D, H, S$ $p \downarrow H, S, D$
- For inferior good (left to right / low to high): $p \uparrow S, H, D$, $p \downarrow D, S, H$

See graphs on the following slides.

Walrasian, Hicksian and Slutsky Demand Curves: Normal Good

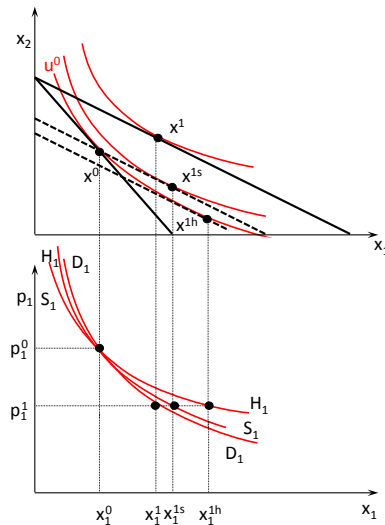


Slutsky demand touching Hicksian demand in x^0 , otherwise above

Walrasian demand below both Slutsky and Hicksian for $p \uparrow$, above for $p \downarrow$

I don't want you to memorize these graphs. I just want you to understand it!

Walrasian, Hicksian and Slutsky Demand Curves: Inferior Good



Slutsky demand touching Hicksian demand in x^0 , otherwise below

Walrasian demand above both Slutsky and Hicksian for $p \uparrow$, below for $p \downarrow$

Index Numbers

Prices change in time. Are consumers better or worse off after the change?

We could use **index numbers** to answer this question.

Let's consider two periods: **base period b** and **current period t**

Accordingly, we denote the corresponding consumption bundles and price vectors as follows:

$$x^b = (x_1^b, x_2^b, \dots, x_n^b), p^b = (p_1^b, p_2^b, \dots, p_n^b)$$

$$x^t = (x_1^t, x_2^t, \dots, x_n^t), p^t = (p_1^t, p_2^t, \dots, p_n^t)$$

Index numbers represent the ratio of the weighted average (of quantities or prices) in the current period (t-period) related to the weighted average (of quantities or prices) in the base period (b-period).

I don't want you to memorize these graphs. I just want you to understand it!

Indices let us evaluate some economic changes based on information about prices and consumption bundles.

We can evaluate changes in cases when BOTH prices and income change.

x^b is a bundle chosen in the base period, x^t is a bundle chosen in the period t

Quantity Index

When we relate the weighted averages of quantities we call the resulting index **quantity index**:

$$I_q = \frac{\sum_{i=1}^n w_i x_i^t}{\sum_{i=1}^n w_i x_i^b}$$

where the weights w_i could be either the current-period prices p_i^t or the base period prices p_i^b :

- Paasche Quantity Index: $P_q = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^t x_i^b}$
- Laspeyres Quantity Index: $L_q = \frac{\sum_{i=1}^n p_i^b x_i^t}{\sum_{i=1}^n p_i^b x_i^b}$

Quantity Indices and Evaluation

Paasche Quantity Index: $P_q = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^t x_i^b}$

- 1 $P_q > 1 \iff \sum_{i=1}^n p_i^t x_i^t > \sum_{i=1}^n p_i^t x_i^b \iff x^t \succ^D x^b$
(x^b available at t , but not chosen $\rightarrow$ consumer is better-off at t)
- 2 $P_q < 1 \iff \sum_{i=1}^n p_i^t x_i^t < \sum_{i=1}^n p_i^t x_i^b$
(x^b unaffordable at $t \rightarrow$ no information about change of welfare in time)

Laspeyres Quantity Index: $L_q = \frac{\sum_{i=1}^n p_i^b x_i^t}{\sum_{i=1}^n p_i^b x_i^b}$

- 1 $L_q < 1 \iff \sum_{i=1}^n p_i^b x_i^t < \sum_{i=1}^n p_i^b x_i^b \iff x^b \succ^D x^t$
(x^t available at b , but not chosen $\rightarrow$ consumer is worse-off at t)
- 2 $L_q > 1 \iff \sum_{i=1}^n p_i^b x_i^t > \sum_{i=1}^n p_i^b x_i^b$
(x^t unaffordable at $b \rightarrow$ no information about change of welfare in time)

x_i^t – quantity of good i consumed in the current period t
 x_i^b – quantity of good i consumed in the base period

- Paasche quantity index – weights are new prices
- Laspeyres quantity index – weights are old prices
- How to remember: L comes before P, so L old (base) prices, P new :-)

How can we evaluate whether the consumer is better off now or in the base period?

It's always about whether the other bundle was or was not available during one of the periods.

- If was, then we can say which is better and therefore in which period consumer is better off
- If was not, then we cannot compare these two periods
- $\sum_{i=1}^n p_i^t x_i^t$ is your spending in period t = income in period t
- If $P_q > 1$, initial bundle x^b lies inside the current budget set (affordable now).
- If $P_q < 1$, initial bundle x^b lies outside the current budget set (not affordable now), but we cannot easily evaluate whether the consumer is better or worse off, we do not know shape of indifference curves.
- $\sum_{i=1}^n p_i^b x_i^b$ is your spending in period b = income in period b
- If $L_q < 1$, current bundle x^t lies inside the old budget set (so it was affordable in the past, but not chosen).
- If $L_q > 1$, current bundle x^t lies outside the old budget set (so it was not affordable in the past), but we cannot evaluate whether the consumer is better or worse off, we do not know shape of indifference curves.

Price Index

When we relate the weighted averages of prices we call the resulting index **price index**:

$$I_p = \frac{\sum_{i=1}^n w_i p_i^t}{\sum_{i=1}^n w_i p_i^b}$$

The related averages of the prices are weighted either by the current-period quantities x_i^t or by the base-period quantities x_i^b :

- Paasche Price Index: $P_p = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^b x_i^t}$
- Laspeyres Price Index: $L_p = \frac{\sum_{i=1}^n p_i^t x_i^b}{\sum_{i=1}^n p_i^b x_i^b}$
- Money Income Index: $M = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^b x_i^b}$

Similar to previous indices, but this time x are fixed and p change.

Money income index measures how income changed between the two periods, $M = \frac{m^t}{m^b}$

If we have different prices in the index, we cannot evaluate the change in welfare. We will assess welfare changes by comparing Laspeyres and Paasche Price indices with Money Income Index (see the next slide).

Price Indices and Evaluation

Analogical to the previous case

$$P_p = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^b x_i^t} \quad L_p = \frac{\sum_{i=1}^n p_i^t x_i^b}{\sum_{i=1}^n p_i^b x_i^b} \quad M = \frac{\sum_{i=1}^n p_i^t x_i^t}{\sum_{i=1}^n p_i^b x_i^b}$$

- ① $P_p > M \iff \sum_{i=1}^n p_i^b x_i^b > \sum_{i=1}^n p_i^b x_i^t \iff x^b \succ^D x^t$
(x^t available at b , but not chosen $\rightarrow$ consumer is worse-off at t)
- ② $L_p < M \iff \sum_{i=1}^n p_i^t x_i^b < \sum_{i=1}^n p_i^t x_i^t \iff x^t \succ^D x^b$
(x^b available at t , but not chosen $\rightarrow$ consumer is better-off at t)

If $P_p < M$ or $L_p > M$, we cannot evaluate the change in consumer's welfare (because x^t was not available at period b or vice versa)

Application: Indexing Social Security Payments

Elderly people, unemployed, handicapped and other socially disadvantaged groups have social security payments as their sole source of income.

Changes in price indices are sometimes used to adjust transfer payments to keep purchasing power constant when prices change. This is called “**indexation**”.

Full indexation – social payments are increased at the same rate as the price index being used to measure the aggregate inflation rate.

In the Czech Republic:

- $CPI = \frac{\sum_i p_i^t w_i^b}{\sum_i p_i^b w_i^b}$, where w_i^b is a weight for good i in basket of consumption commodities (weights adjusted every 2 years)
- pensions for retirees are fully indexed by CPI and partly by growth rate of real wage
- retirees are always weakly better off after the price increase (and corresponding indexation) if their weights on goods in their consumption basket are similar to w^b

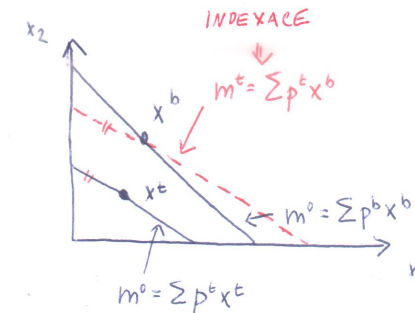
- Original consumption x^b , original income $m^0 = \sum_i p_i^b x_i^b$
- Prices increased from p^b to p^t
- Consumption before indexation x^t , income as in period b : $m^0 = \sum_i p_i^t x_i^t$
- If x^t is chosen in the current period, when consumer has the same income as before, but prices are higher, x^t must have been available also in the base period, i.e. $\sum_i p_i^b x_i^t < \sum_i p_i^b x_i^b$ and $x^b \succ x^t$
- The original income m^0 will be indexed by CPI.
- $CPI = \frac{\sum_i p_i^t w_i^b}{\sum_i p_i^b w_i^b}$

What are weights in CPI?

- Average household: $\sum_i p_i x_i = m$, we can rewrite it as $\sum_i p_i \frac{x_i}{m} = 1$ and weights $w_i^b = \frac{1}{m} x_i^b = k \cdot x_i^b$, k is constant
- Weight expresses how many units of x_i are bought on average from 1 dollar of income

After indexation:

$$m^t = CPI \cdot m^0 = \frac{\sum_i p_i^t w_i^b}{\sum_i p_i^b w_i^b} \cdot m^0 = \frac{k \sum_i p_i^t x_i^b}{k \sum_i p_i^b x_i^b} \cdot \sum_i p_i^b x_i^b = \sum_i p_i^t x_i^b$$



Weakly better off

- Because they can substitute to get better, if not, they will keep the consumption the same and stay at the same utility level.

However, if retirees' weights are very different, they may not be better off as the price of the good they consume may rise substantially and only get a small share of the indexation basket. So they will not be able to buy the original bundle after the indexation.